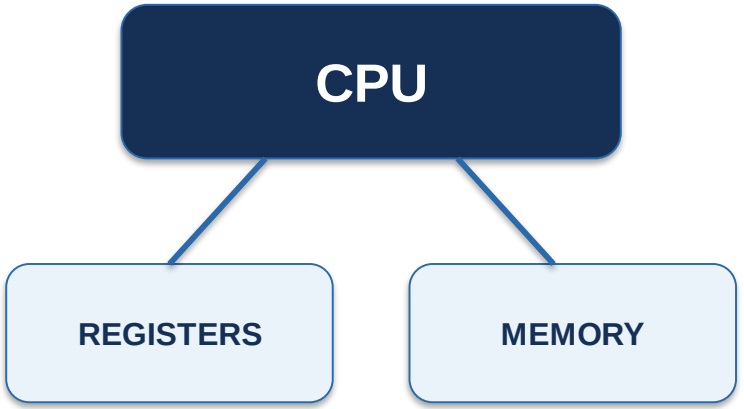
Where the Machine Keeps Things



Registers, memory and the path from an address to stored data.

- Registers
- Register file
- RAM & ROM
- Memory hierarchy

Concepts → architecture → working computer

1. Why does a CPU need storage?

Registers and memory hold the values that instructions use.

Registers

Very small, very fast storage inside the CP
Hold values being used right now.

Main memory (RAM)

Stores the program and data currently in u
Larger than registers, but slower.

Read-only memory (ROM)

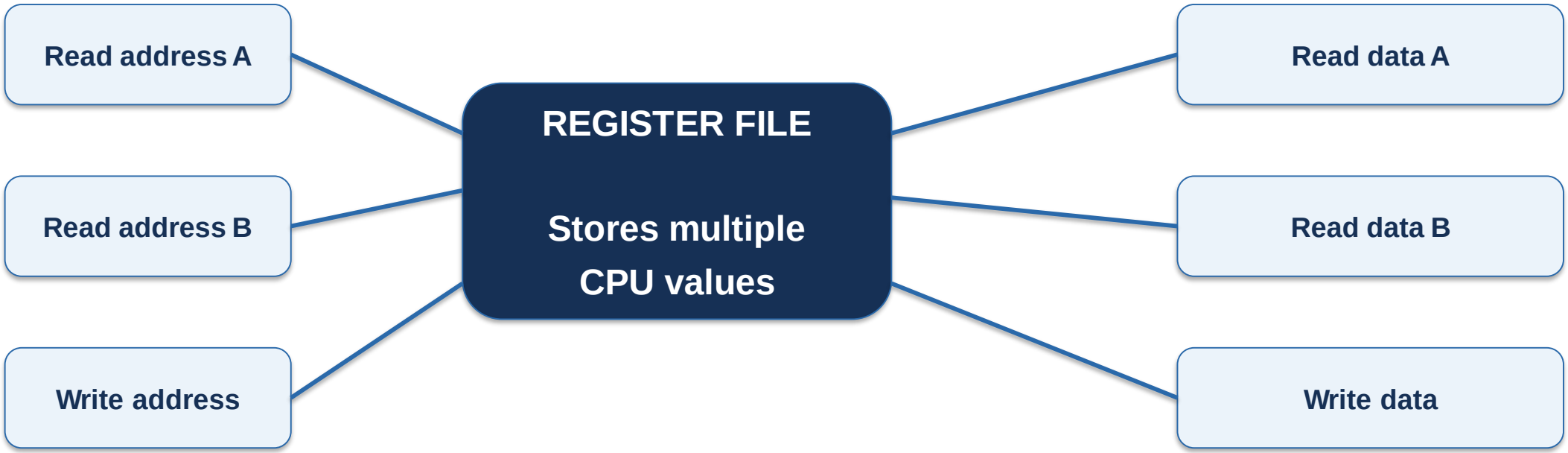
Stores fixed instructions or constants in syste

CPU registers — **Main memory** — **Long-term storage**

Key idea: the closer storage is to the CPU, the faster and smaller it usually is.

2. The register file

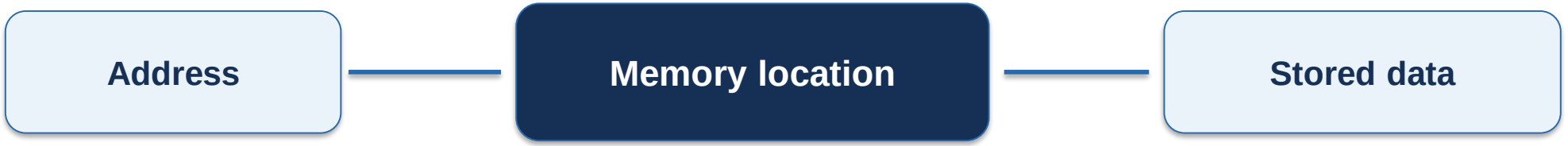
A collection of CPU registers that instructions can read and write.



A typical register file can provide two values to an instruction and write one result back.

3. How memory is organised

Addresses identify locations; data is the information stored there.



Read

Give memory an address → memory returns the data sto

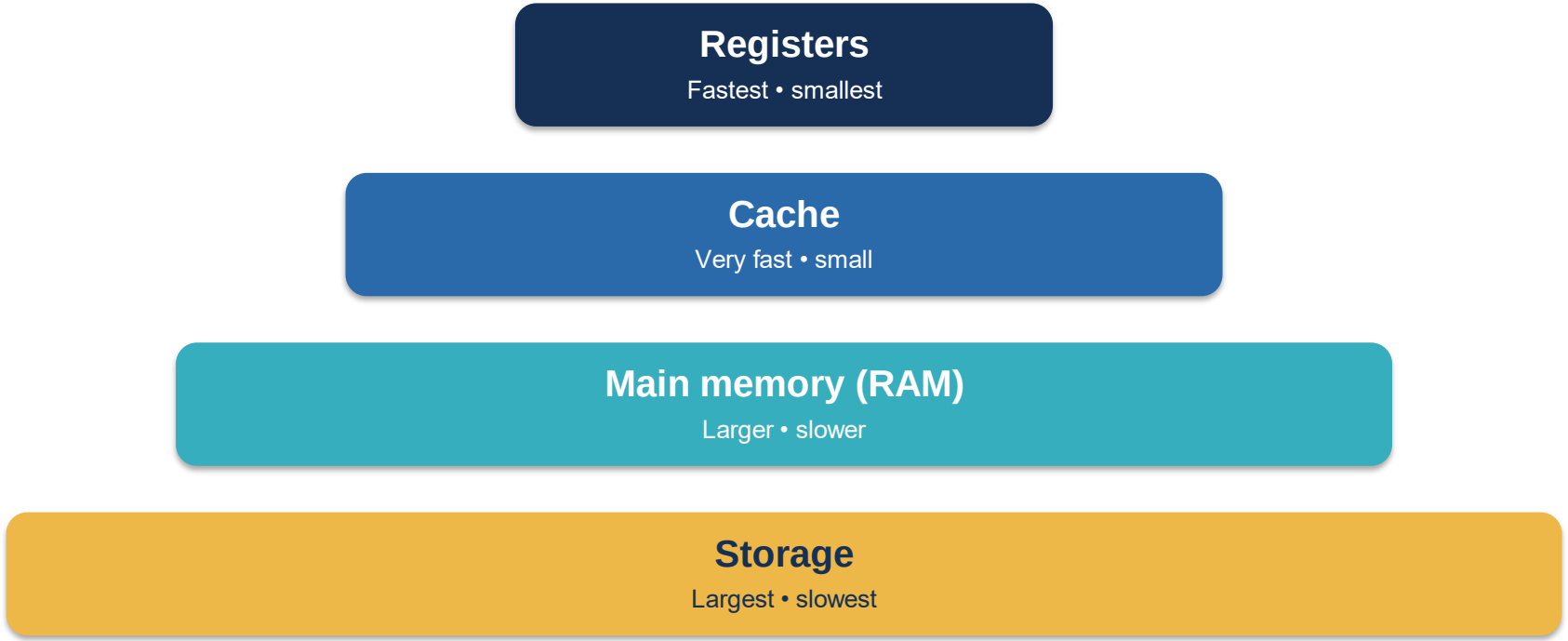
Write

Give an address and data → memory stores that data at the locat

Memory is organised into locations. Each location has an address, and each address refers to stored information.

4. Why do we need a memory hierarchy?

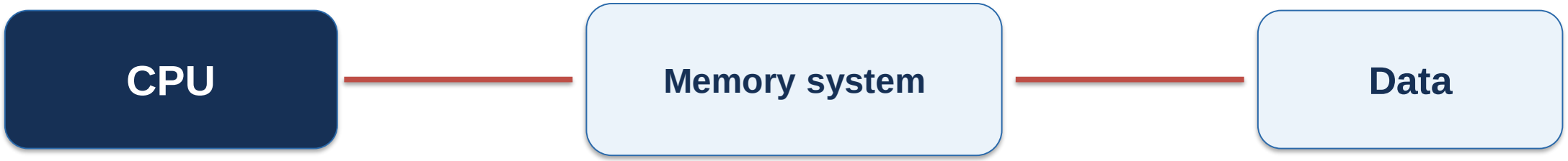
Different storage levels balance speed, size and cost.



The hierarchy helps the CPU access frequently used information quickly without making all storage expensive.

5. The memory bottleneck

The CPU can be fast, but it still has to wait for data.



Why it matters

CPU performance can be limited by the rate at which instruc

One way to help

Caches keep copies of frequently used data closer to the CPU.